Complete Formula Sheet

38 GHz 8-Point Star-Shaped 2×1 Microstrip Antenna Array

Analytical Design, CST Geometry, Matching and Performance Equations

Design basis. The equations below include the analytical starting equations and the formulas used to define, tune and verify the CST model. Because the final radiator is an 8-point star rather than a conventional rectangular patch, rectangular-patch equations are used as **initial sizing references**; the final star dimensions and feed were then refined through full-wave CST simulation.

Final Design Parameters Used in CST

Parameter	Value	Parameter	Value
Operating frequency, f_0	38 GHz	Relative permittivity, ε_r	2.2
Substrate	Rogers RT/duroid 5880	Substrate thickness, h	0.254 mm
Copper thickness, t_{cu}	0.035 mm	Board width, W_{sub}	10.5 mm
Board length, L_{sub}	6.5 mm	Array spacing, d_{array}	3.947 mm
Base inner radius, R_{in0}	1.229 mm	Base outer radius, R_{out0}	1.707 mm
Final 50-Ω feed width	about 0.70 mm	Matching length used	about 0.62 mm

1. Basic Frequency, Units and Wavelength

Frequency conversion

$$\boxed{f(\text{Hz}) = f(\text{GHz}) \times 10^9}$$

For this design:

$$38\ \text{GHz} = 38 \times 10^9\ \text{Hz}$$

Length conversions

$$\boxed{1\ \text{m} = 1000\ \text{mm}}, \qquad \boxed{1\ \text{mm} = 10^{-3}\ \text{m}}, \qquad \boxed{1\ \mu\text{m} = 10^{-3}\ \text{mm}}$$

Speed of light

$$\boxed{c \approx 3 \times 10^8\ \text{m/s}}$$

Free-space wavelength

$$\boxed{\lambda_0 = \frac{c}{f_0}}$$

At 38 GHz:

$$\lambda_0 = \frac{3 \times 10^8}{38 \times 10^9} = 7.895\ \text{mm (approximately)}$$

$$\boxed{\lambda_0 \approx 7.895\ \text{mm}}$$

Half wavelength and quarter wavelength

$$\boxed{\frac{\lambda_0}{2} \approx 3.947\ \text{mm}}, \qquad \boxed{\frac{\lambda_0}{4} \approx 1.974\ \text{mm}}$$

Free-space wave number

$$\boxed{k_0 = \frac{2\pi}{\lambda_0} = \frac{2\pi f_0}{c}}$$

2. Conventional Rectangular-Patch Reference Equations

Initial patch width

$$W = \frac{c}{2f_0}\sqrt{\frac{2}{\varepsilon_r + 1}}$$

Effective dielectric constant

$$\varepsilon_{eff} = \frac{\varepsilon_r + 1}{2} + \frac{\varepsilon_r - 1}{2}\left(1 + \frac{12h}{W}\right)^{-1/2}$$

Fringing-field length extension

$$\frac{\Delta L}{h} = 0.412\frac{(\varepsilon_{eff} + 0.3)\left(\frac{W}{h} + 0.264\right)}{(\varepsilon_{eff} - 0.258)\left(\frac{W}{h} + 0.8\right)}$$

Hence,

$$\Delta L = 0.412h\frac{(\varepsilon_{eff} + 0.3)\left(\frac{W}{h} + 0.264\right)}{(\varepsilon_{eff} - 0.258)\left(\frac{W}{h} + 0.8\right)}$$

Effective and physical patch length

$$L_{eff} = \frac{c}{2f_0\sqrt{\varepsilon_{eff}}}$$

$$L = L_{eff} - 2\Delta L$$

3. 8-Point Star Geometry

Number of vertices

For an 8-point star, the polygon alternates between 8 outer and 8 inner vertices:

$$N_{outer} = 8, \qquad N_{inner} = 8, \qquad N_{total} = 16$$

Angular spacing

$$\Delta\theta = \frac{360^\circ}{16} = 22.5^\circ$$

In radians:

$$\Delta\theta = \frac{2\pi}{16} = \frac{\pi}{8}$$

Scaled inner and outer radius

$$R_{in} = R_{in0}S_{star}, \qquad R_{out} = R_{out0}S_{star}$$

with

$$R_{in0} = 1.229\,\text{mm}, \qquad R_{out0} = 1.707\,\text{mm}.$$

Star vertex coordinates

For vertex i:

$$\boxed{x_i = x_c + r_i \cos\theta_i}$$

$$\boxed{y_i = y_c + r_i \sin\theta_i}$$

where

$$\boxed{\theta_i = i\frac{\pi}{8}}, \qquad i = 0, 1, 2, \ldots, 15$$

and

$$r_i = \begin{cases} R_{out}, & i \text{ even}, \\ R_{in}, & i \text{ odd}. \end{cases}$$

Star diameters

$$\boxed{D_{out} = 2R_{out}}, \qquad \boxed{D_{in} = 2R_{in}}$$

Approximate resonance scaling relation

If the whole radiator is scaled uniformly:

$$\boxed{f_r \propto \frac{1}{L_{electrical}}}$$

A useful tuning estimate is:

$$\boxed{S_{new} \approx S_{old}\frac{f_{old}}{f_{target}}}$$

Increasing S_{star} makes the star electrically larger and generally shifts resonance lower; decreasing it shifts resonance higher.

4. 2×1 Array Relations

Element spacing

A common first choice is half a free-space wavelength:

$$\boxed{d_{array} \approx \frac{\lambda_0}{2}}$$

At 38 GHz:

$$\boxed{d_{array} \approx 3.947\,\text{mm}}$$

Symmetric element positions

$$\boxed{x_1 = -\frac{d_{array}}{2}}, \qquad \boxed{x_2 = +\frac{d_{array}}{2}}$$

Numerically:

$$x_1 = -1.9735\ \text{mm}, \qquad x_2 = +1.9735\ \text{mm}.$$

Verification:

$$\boxed{x_2 - x_1 = d_{array}}$$

Two-element array factor

General form:

$$AF = 2\cos\left(\frac{kd\sin\theta + \beta}{2}\right)$$

where

$$k = \frac{2\pi}{\lambda}, \quad d = \text{element spacing}, \quad \beta = \text{progressive phase}.$$

For broadside excitation, $\beta = 0$:

$$AF = 2\cos\left(\frac{kd\sin\theta}{2}\right)$$

5. Microstrip Feed-Line Equations

Normalized width

$$u = \frac{W_f}{h}$$

Effective permittivity of a microstrip line

A widely used approximation is:

$$\varepsilon_{eff} \approx \frac{\varepsilon_r + 1}{2} + \frac{\varepsilon_r - 1}{2}\left(1 + \frac{12h}{W_f}\right)^{-1/2}$$

Characteristic impedance for narrow microstrip

For approximately $W_f/h \le 1$:

$$Z_0 = \frac{60}{\sqrt{\varepsilon_{eff}}}\ln\left(\frac{8h}{W_f} + \frac{W_f}{4h}\right)$$

Characteristic impedance for wider microstrip

For approximately $W_f/h \ge 1$:

$$Z_0 = \frac{120\pi}{\sqrt{\varepsilon_{eff}}\left[\frac{W_f}{h} + 1.393 + 0.667\ln\left(\frac{W_f}{h} + 1.444\right)\right]}$$

Guided wavelength

$$\lambda_g = \frac{\lambda_0}{\sqrt{\varepsilon_{eff}}}$$

6. T-Junction and Quarter-Wave Matching

Parallel impedance

$$Z_{eq} = Z_1 \parallel Z_2 = \frac{Z_1 Z_2}{Z_1 + Z_2}$$

For equal branches, $Z_1 = Z_2 = 100\ \Omega$:

$$Z_{eq} = \frac{100 \times 100}{100 + 100} = 50\ \Omega.$$

Quarter-wave transformer impedance

$$Z_T = \sqrt{Z_0 Z_L}$$

For $Z_0 = 50\ \Omega$ and $Z_L = 100\ \Omega$:

$$Z_T = \sqrt{50 \times 100} = 70.71\ \Omega$$

Quarter-wave transformer length

$$L_T = \frac{\lambda_g}{4} = \frac{\lambda_0}{4\sqrt{\varepsilon_{eff}}}$$

7. CST Parametric Geometry Relations

Matching-section vertical position

The design used the parametric relationship:

$$y_{match} = y_{star} - R_{in} + L_{overlap} - L_t$$

Feed overlap with the radiator

$$y_{feed,max} = y_{star} - R_{in} + L_{overlap}$$

Substrate X and Y limits

$$x_{min} = -\frac{W_{sub}}{2}, \qquad x_{max} = +\frac{W_{sub}}{2}$$

$$y_{min} = -\frac{L_{sub}}{2}, \qquad y_{max} = +\frac{L_{sub}}{2}$$

8. Layer Stack-Up and Z Coordinates

Ground plane

$$z_{ground,min} = 0, \qquad z_{ground,max} = t_{cu}$$

For $t_{cu} = 0.035$ mm:

$$0 \rightarrow 0.035 \text{ mm}$$

Substrate

$$z_{sub,min} = t_{cu}, \qquad z_{sub,max} = t_{cu} + h$$

Numerically:

$$0.035 \text{ mm} \rightarrow 0.289 \text{ mm}$$

Thickness check:

$$0.289 - 0.035 = 0.254 \text{ mm}$$

Top patch and feed copper

$$z_{top,min} = t_{cu} + h$$

$$z_{top,max} = h + 2t_{cu}$$

Numerically:

$$0.289 \text{ mm} \rightarrow 0.324 \text{ mm}$$

Overall physical thickness

$$T_{total} = t_{cu} + h + t_{cu} = h + 2t_{cu}$$

$$T_{total} = 0.254 + 2(0.035) = 0.324 \text{ mm}$$

9. Reflection Coefficient, S11 and Return Loss

Reflection coefficient

$$\boxed{\Gamma = \frac{Z_{in} - Z_0}{Z_{in} + Z_0}}$$

Perfect matching gives $\Gamma = 0$; complete reflection gives $|\Gamma| = 1$.

S11 in dB

$$\boxed{S_{11}(\text{dB}) = 20 \log_{10} |\Gamma|}$$

Reflection coefficient from S11

$$\boxed{|\Gamma| = 10^{S_{11}(\text{dB})/20}}$$

Return loss

$$\boxed{RL = -20 \log_{10} |\Gamma|}$$

When S11 is plotted as a negative dB value:

$$\boxed{RL = -S_{11}(\text{dB})}$$

10. VSWR

VSWR from reflection coefficient

$$\boxed{VSWR = \frac{1 + |\Gamma|}{1 - |\Gamma|}}$$

Reflection coefficient from VSWR

$$\boxed{|\Gamma| = \frac{VSWR - 1}{VSWR + 1}}$$

Ideal matching gives $VSWR = 1$.

11. Input Impedance

Complex impedance

$$\boxed{Z_{in} = R + jX}$$

At resonance, ideally:

$$\boxed{X \approx 0}$$

For a nominal 50-Ω system:

$$\boxed{Z_{in} \approx 50 + j0\ \Omega}$$

Impedance magnitude

$$\boxed{|Z| = \sqrt{R^2 + X^2}}$$

Impedance phase

$$\boxed{\angle Z = \tan^{-1}\left(\frac{X}{R}\right)}$$

12. Bandwidth and Quality Factor

-10 dB bandwidth

For the final approved design:

$$f_L \approx 37.504\,\text{GHz}, \qquad f_H \approx 38.520\,\text{GHz}.$$

$$\boxed{BW = f_H - f_L}$$

$$BW = 38.520 - 37.504 = \boxed{1.016\,\text{GHz}}$$

Band-centre frequency

$$\boxed{f_c = \frac{f_H + f_L}{2}}$$

Fractional bandwidth

$$\boxed{FBW = \frac{BW}{f_0} \times 100\%}$$

Using $BW = 1.016$ GHz and $f_0 = 38$ GHz:

$$\boxed{FBW \approx 2.67\%}$$

Approximate quality factor

$$\boxed{Q \approx \frac{f_0}{BW}}$$

13. Power and Mismatch Relations

Reflected power

$$\boxed{P_{reflected} = P_{inc}|\Gamma|^2}$$

Accepted power

$$\boxed{P_{accepted} = P_{inc}(1 - |\Gamma|^2)}$$

Mismatch efficiency

$$\boxed{\eta_m = 1 - |\Gamma|^2}$$

Mismatch loss

$$\boxed{ML = -10\log_{10}(1 - |\Gamma|^2)}$$

14. Radiation Efficiency and Total Efficiency

Radiation efficiency

$$\boxed{\eta_{rad} = \frac{P_{rad}}{P_{accepted}}}$$

Percentage form:

$$\boxed{\eta_{rad}(\%) = 100\eta_{rad}}$$

Total efficiency

$$\eta_{tot} = \eta_{rad}\eta_m$$

Therefore:

$$\eta_{tot} = \eta_{rad}(1 - |\Gamma|^2)$$

Efficiency conversion from dB

$$\eta_{linear} = 10^{\eta_{dB}/10}$$

$$\eta(\%) = 100 \times 10^{\eta_{dB}/10}$$

15. Directivity, Gain and Effective Aperture

Directivity

$$D = \frac{4\pi U_{max}}{P_{rad}}$$

Directivity in dBi

$$D_{dBi} = 10\log_{10} D$$

Gain

$$G = \eta_{rad} D$$

If mismatch is included:

$$G_{realized} = \eta_{tot} D$$

Gain in dBi

$$G_{dBi} = 10\log_{10} G$$

Linear gain from dBi

$$G_{linear} = 10^{G_{dBi}/10}$$

Effective aperture

$$A_e = \frac{G\lambda_0^2}{4\pi}$$

where G is in linear form.

16. Radiation Pattern Quantities

Half-power beamwidth

The half-power points occur at approximately -3 dB relative to the main-lobe peak.

$$HPBW = |\theta_2 - \theta_1|$$

Main-lobe direction

For the final broadside design:

$$\theta_{max} \approx 0^\circ$$

Side-lobe level

$$SLL = G_{sidelobe,max} - G_{main,max}$$

The result is normally a negative dB value.

17. Far-Field Distance

A commonly used criterion is:

$$R_{FF} \geq \frac{2D^2}{\lambda_0}$$

where D is the largest physical dimension of the radiating structure.

18. Final Guide-Approved CST Results

f_0	$= 38\,\text{GHz}$
$S_{11}(38\ \text{GHz})$	$\approx -29.464\ \text{dB}$
$VSWR$	≈ 1.0705
Z_{ref}	$\approx 50.388 + j0\ \Omega$
$BW_{-10\,dB}$	$\approx 1.016\,\text{GHz}$
FBW	$\approx 2.67\%$
$Gain_{3D}$	$\approx 6.481\ \text{dBi}$
$Radiation\ efficiency$	$\approx 89.36\%$
$Total\ efficiency$	$\approx 89.26\%$
$HPBW\ (\phi = 0^\circ)$	$\approx 48.3^\circ$
$Main\ lobe\ direction$	$\approx 0^\circ$
$Side\ lobe\ level$	$\approx -19.3\ \text{dB}$

19. Quick Viva Formula List

Quantity	Formula
Wavelength	$\lambda_0 = \dfrac{c}{f}$
Array spacing	$d \approx \dfrac{\lambda_0}{2}$
Effective permittivity	$\varepsilon_{eff} = \dfrac{\varepsilon_r + 1}{2} + \dfrac{\varepsilon_r - 1}{2}\left(1 + \dfrac{12h}{W}\right)^{-1/2}$
Quarter-wave transformer	$Z_T = \sqrt{Z_0 Z_L}$
Transformer length	$L_T = \dfrac{\lambda_g}{4}$
Star scaling	$R_{in} = R_{in0} S_{star}, \quad R_{out} = R_{out0} S_{star}$
Reflection coefficient	$\Gamma = \dfrac{Z_{in} - Z_0}{Z_{in} + Z_0}$
S11	$S_{11} = 20 \log_{10} \lvert\Gamma\rvert$
VSWR	$VSWR = \dfrac{1 + \lvert\Gamma\rvert}{1 - \lvert\Gamma\rvert}$
Bandwidth	$BW = f_H - f_L$
Fractional bandwidth	$FBW = \dfrac{BW}{f_0} \times 100\%$
Gain	$G = \eta D$
Total efficiency	$\eta_{tot} = \eta_{rad}(1 - \lvert\Gamma\rvert^2)$
Far-field distance	$R_{FF} \geq \dfrac{2D^2}{\lambda_0}$

Prepared as a study, design and viva reference for the 38 GHz 8-point star-shaped 2×1 microstrip antenna array.